# Julien Laurendeau

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## Education

- Ph.D in Mathematics, École Polytechnique Fédérale de Lausanne (EPFL), 2023–present
  • Thesis advisor: Prof. Mats Stensrud, Chair of Biostatistics
- M.Sc. in Mathematics, with distinction, ETH Zürich, 2021–2023
  • Master's thesis: Quantile Treatment Effects, supervised by Prof. Nicolaï Meinshausen
- B.Sc. in Mathematics, École Polytechnique Fédérale de Lausanne (EPFL), 2018–2021
  • Bachelor's thesis: The Generalized Minimum Manhattan Network Problem, supervised by Dr. Martina Gallato and Prof. Friedrich Eisenbrand, Chair of Discrete Optimization

## Research Experience

- Summer 2022: Summer Intern, Chair of Biostatistics, EPFL
  • Supervisor: Prof. Mats Stensrud
- 2022–2023: Research Assistant, Faculty of Macroeconomics, ETH Zürich
  • Supervisor: Prof. Hans Gersbach

## Professional Service

- Reviewer, Journal of the American Statistical Association, 2024–2026
- Reviewer, Journal of Machine Learning Research, 2024–2026

## Awards

- Early Career Poster Competition Winner, European Causal Inference Meeting, University of Oxford, April 2026

## Languages

- English (native), French (native), German (conversational)

## Publications

- Julien D. Laurendeau, Aaron L. Sarvet, Mats J. Stensrud. (2025). "Improved bounds and inference on optimal regimes." Journal of the American Statistical Association — Theory and Methods, 121(553), 561–573.
- Aaron L. Sarvet, Julien D. Laurendeau, Mats J. Stensrud. (2024). "Optimal regimes with limited resources." arXiv:2403.19842 [stat.ME].
- Mats J. Stensrud, Julien D. Laurendeau, Aaron L. Sarvet. (2024). "Optimal regimes for algorithm-assisted human decision-making." Biometrika, 111(4), 1089–1108.

## Talks

- New guarantees for optimal regimes in the presence of unmeasured confounding — Invited talk at 7th Meeting of the Institute of Mathematical Statistics Asia Pacific Rim Meeting (IMS-APRM), Chinese

University of Hong Kong, Hong Kong, 2026
- Finding treatment regimes with optimality guarantees of practical interest — Poster at European Causal Inference Meeting, University of Oxford, Oxford, United Kingdom, 2026
- Optimal sequential decision-making with initiation regimes — Poster at Foundations of Causal Inference Workshop, part of the Causal inference: From theory to practice and back again programme, Isaac Newton Institute for Mathematical Sciences, Cambridge, United Kingdom, 2026
- Optimal sequential decision-making with initiation regimes — Poster at Causal Machine Learning for the Social Sciences Workshop, part of the Causal inference: From theory to practice and back again programme, Isaac Newton Institute for Mathematical Sciences, Cambridge, United Kingdom, 2026
- Optimal sequential decision-making with initiation regimes — Contributed talk at 46th Annual Conference of the International Society for Clinical Biostatistics (ISCB), Biozentrum of University of Basel and Department of Biosystems Science and Engineering (D-BSSE) of ETH Zürich, Basel, Switzerland, 2025
- Optimal sequential decision-making with initiation regimes — Poster at European Causal Inference Meeting, Ghent University, Ghent, Belgium, 2025
- Optimal sequential decision-making with initiation regimes — Contributed flash talk at Causal Inference in the Health Sciences Symposium, University of Zurich, Zurich, Switzerland, 2025
- Optimal sequential decision-making with initiation regimes — Lightning talk and poster at Swiss Statistics Seminar, Universität Bern, Bern, Switzerland, 2025
- New guarantees for optimal regimes in the presence of unmeasured confounding — Contributed talk at IMS International Conference on Statistics and Data Science, Nice, France, 2024
- Improved bounds and inference on optimal regimes — Contributed talk at European Causal Inference Meeting, University of Copenhagen, Copenhagen, Denmark, 2024
- Improved bounds and inference on optimal regimes — Poster at Causal Inference in the Health Sciences Symposium, EPFL, Lausanne, Switzerland, 2024

## Teaching

- Master's thesis supervision: "Monotonicity and the superoptimal regime," EPFL, Spring Semester 2026.
- Head Teaching Assistant for Randomization and Causation, EPFL, Spring Semester 2026 (Instructor: Dr. Nils Sturma).
- Semester project supervision: "Testing for effect heterogeneity in causal inference," EPFL, Fall Semester 2025.
- Head Teaching Assistant for Causal Thinking, EPFL, Fall Semester 2025 (Instructor: Prof. Mats Stensrud).
- Semester project supervision: "Superoptimal Markov Decision Processes in Games," EPFL, Spring Semester 2025.
- Head Teaching Assistant for Statistics for Mathematicians, EPFL, Spring Semester 2025 (Instructor: Prof. Victor Panaretos).
- Head Teaching Assistant for Causal Thinking, EPFL, Fall Semester 2024 (Instructor: Prof. Mats Stensrud).
- Master's thesis supervision: "Violations of the Single Unit Treatment Value Assumption," EPFL, Fall Semester 2024.
- Co-Head Teaching Assistant for Applied Biostatistics, EPFL, Spring Semester 2024 (Instructor: Dr. Darlene Goldstein).
- Semester project supervision: "Time-varying optimal regimes," EPFL, Spring Semester 2024.
- Head Teaching Assistant for Probability and Statistics for Physicists, EPFL, Fall Semester 2023

(Instructor: Prof. Anthony Davison).
- Head Teaching Assistant for Analysis II for Life Sciences, EPFL, Spring Semester 2023 (Instructor: Dr. Adélie Garin).
- Teaching Assistant for Analysis IV for Physicists, EPFL, Spring Semester 2021 (Instructor: Prof. Joachim Stubbe).
- Teaching Assistant for Numerical Analysis, EPFL, Spring Semester 2021 (Instructor: Prof. Daniel Kressner).
- Teaching Assistant for Linear Algebra, Cours de Mathématiques Spéciales (CMS), EPFL, Fall Semester 2020 (Instructor: Dr. Guido Burmeister).